

Single Source Justification (Simplified Acquisition)

Section A: General Contract Information

Section B: Description of the Supplies/Services Required

Siemens support policy dictates that third-party vendors are directly responsible for customer support for any products that they sell. Specialized technical support is REQUIRED in areas including, but not limited to: GPGPU execution of the software, the advanced unsteady and multi-disciplinary turbomachinery meshing and simulation and post-processing capabilities, polyhedral and advancing layer mesh generation, harmonic balance, parametric geometry within the specialized Star CAD kernel, advanced processes for tight integration of Star with pre- and post-processing pipelines and additional solvers, massively parallel simulations including thousands of processors on DOD supercomputing resources, and batch processing through advanced Star scripting languages.

Section C: Justification for Soliciting from a Single Source or Brand Name

This justification is prepared IAW RFO 12.102(a).

The purchase of the Siemens STAR-CCM+ Power Session Plus is required to obtain access to GPGPU-enable executables that are 20.2X faster than the CPU-based software. The purchase supports the expanding number of simulations required by the mission of the Turbomachinery Branch of the Air Force Research Labs (AFRL/RQTT), and ensures that AFRL/RQTT remains mission capable. The Siemens STAR-CCM+ Power Session Plus license will provide AFRL/RQTT the ability to use STAR-CCM+ products on local CPU- and GPGPU-based systems for execution of high-priority, large-scale simulations while keeping installations up to date with the latest version.

STAR-CCM+ products used in AFRL/RQTT are for CFD analysis of engine and engine component test configurations for emerging aerospace systems. The products include an entire suite of geometry manipulation; parallel, polyhedral mesh generation; multi-physics problem setup; mixing-plane, harmonic balance, and dual time-stepping simulation of turbomachinery components; massively parallel analysis capability; and extensive co- and post-processing analyses. Additionally, the Power Session Plus license allows solver use on an unlimited number of cores and an unlimited number of GPGPUs by any user through use of a floating license server.

The STAR-CCM+ product is used extensively by AFRL/RQTT to rapidly mesh and analyze unique geometries for inlet

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and turbomachinery components and is required for compatibility with DOD High Performance Computing work and historical solution sets. The software has been internally validated using an extensive experimental database and has been demonstrated to meet the Government's requirements for accuracy on a variety of configurations. STAR-CCM+ is used to analyze complex airframe-propulsion configurations for new platforms being developed internally in RQ and to independently verify the physics on externally designed future aircraft platforms.

Section D: Efforts to Obtain Competition

Market research was performed by AFRL/RQOC GSA Advantage! contracts GS-35F-0696M (STAR-CCM+) and GS-35F-0639N (ANSYS CFD). A quote was obtained from Siemens Government Technologies. In conducting market research for the alternative software products, the prices of parallel and GPGPU execution on the RQTT in-house computing resources were assessed.

AFRL/RQTT has been a leader in GPGPU-based computing for computational fluid dynamics and is very familiar with GPGPU software in academia, industry, and through commercial vendors.

Purchase of software from a supplier also determines what entity will provide software support. Siemens Government Technology offers personalized software maintenance support designed specifically for the Star-CCM+ software and the detailed requirements found in III and IV, a service not available when purchasing from other suppliers. This support by Siemens Government Technology includes but is not limited to areas such as GPGPU computing, turbomachinery, conjugate heat transfer, harmonic balance, gas turbine engine simulation, and integration with other software for multi-domain analyses. Spatial Integrated Systems and other potential vendors only list expertise in the Solid Edge, NX CAD, NX CAM & FEMAP products which are unrelated to the technical requirements for this purchase. Therefore, Siemens Government Technology is the only fully qualified supplier of the Star-CCM+ software maintenance and the only supplier that meets the technical requirements described.

Section E: Steps to Preclude Future Single Source or Brand Name Awards

AFRL/RQTT will continue to monitor the market to determine if other products meet the government's needs. The



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government will continue to conduct market research using catalogs, the internet, attendance at professional and industry events, and other

resources to locate other solutions to meet the government's requirements. Additional software products will be evaluated periodically for capability, ability to meet technical requirements, compatibility with industry, and compatibility with in-house and DOD supercomputing resources.

Section F: Contracting Officer's Determination

I have determined, in accordance with [FAR 13.106-1\(b\)\(1\)](#), that the circumstances of this contract action deem only a single source is reasonably available.

